

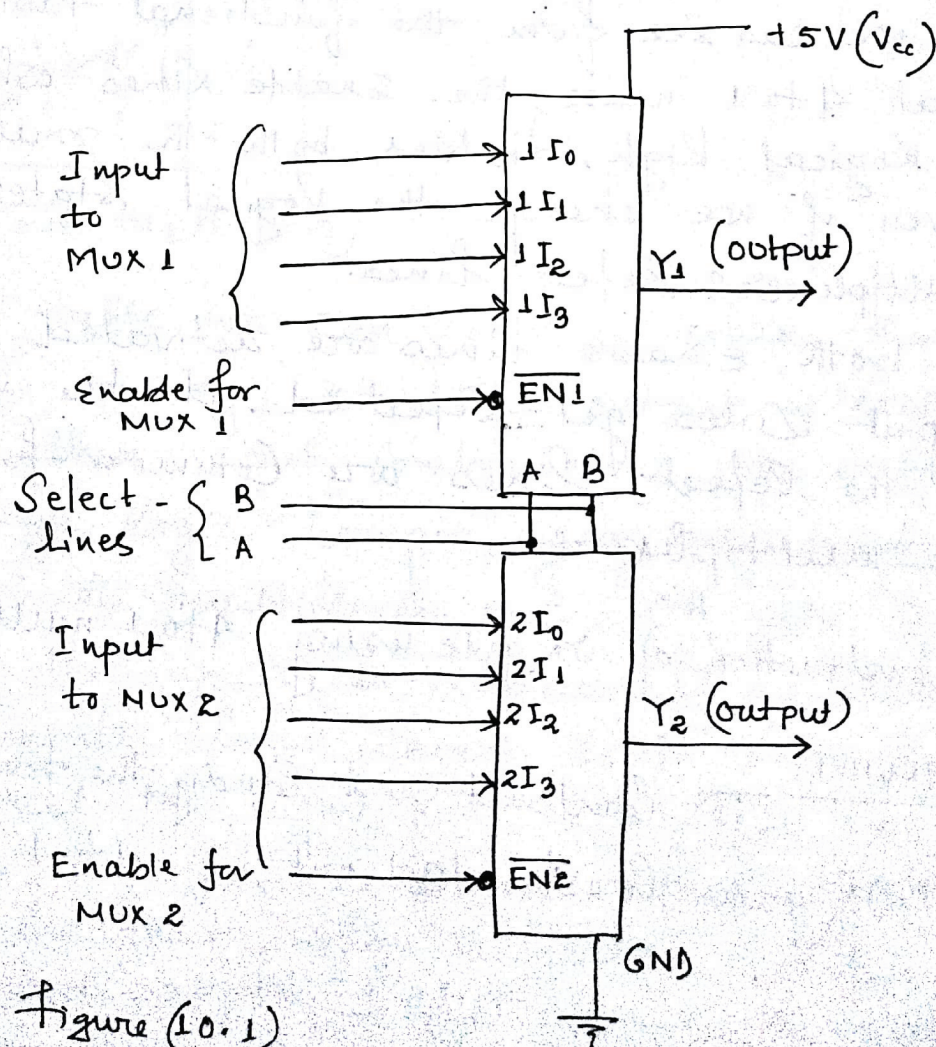
Exp NO.10 Study of multiplexer and construction of logic gates using multiplexer.

(a) OBJECTIVE:

Study the operation characteristics of Dual 4 to 1 multiplexer and construct basic and derived logic gates.

THEORY:

Dual 4 to 1 multiplexer Contains two individual 4 to 1 multiplexer with Common select lines. The TTL integrated ICircuit 74LS153 is a dual 4 to 1 multiplexer. The block diagram of dual 4 to 1 multiplexer has been shown in the figure (10.1).



The functional table of multiplexer has been shown in the truth table (10.1).

Enable line		Select lines		Output	
EN1	EN2	B	A	Y_1	Y_2
H	H	X	X	L	L
L	L	0	0	$1 I_0$	$2 I_0$
L	L	0	1	$1 I_1$	$2 I_1$
L	L	1	0	$1 I_2$	$2 I_2$
L	L	1	1	$1 I_3$	$2 I_3$

Truth table (10.1)

As we can see from the functional table of the dual 4 to 1 mux, the enable lines when connected to logical high, disables both the multiplexer even if we change the logical states of multiplexers select lines.

If both enable lines are activated the input lines get selected at the output as the select lines are common for both the multiplexers.

(b) Construction of OR gate using 4 to 1 multiplexer.

OBJECTIVE!

To Construct and Study the truth table of OR gate Constructed out of 4 to 1 multiplexer.

THEORY!

OR gate is one of basic gate, which is functionally incomplete and is capable of performing only one of the Boolean functions i.e. ORing.

We can construct a two input OR gate using available Dual 4 to 1 multiplexer.

The truth/functional table of OR gate is given in truth table (10.2).

I_0	A	Y
0	0	0
0	1	1
1	0	1
1	1	1

Truth table (10.2)

The Boolean expression for OR gate is as follows

$$Y = A + B$$

- From the table we can see that when the inputs are at logic low, the output is low and for other input combinations the output is at high logic. Comparing or augmenting the table of multiplexer with OR gate we can set the inputs as desired. If we connect all the inputs lines except I_0 to logic high and connect the input line I_0 to logic low and use the select lines as inputs of OR gate, we can see that, when different combinations are provided at the select lines, the output

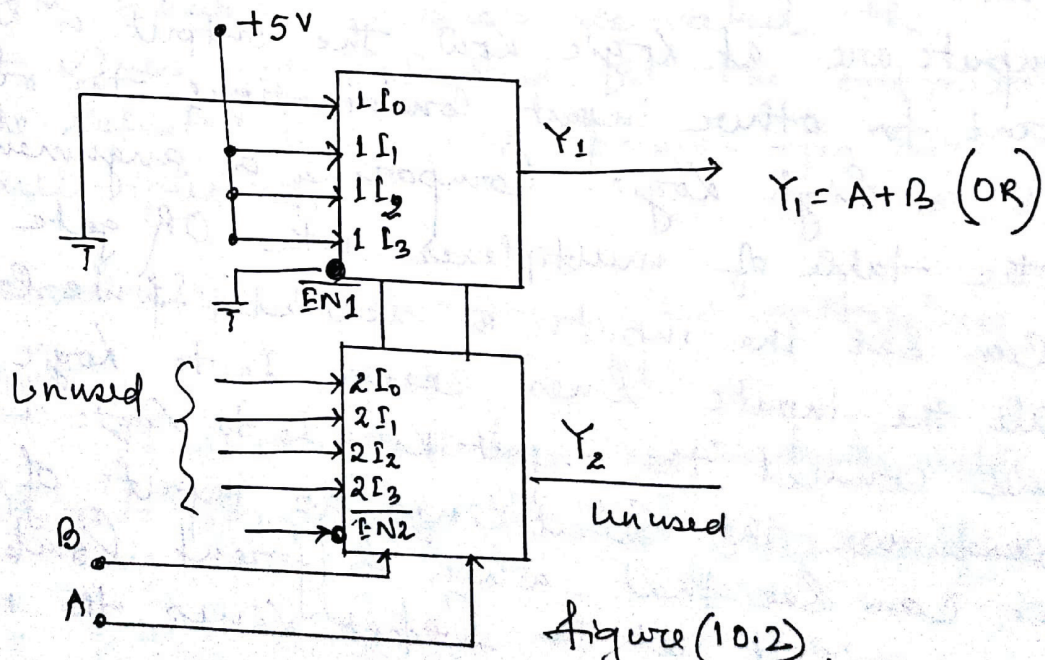
of the multiplexer behaves as OR gate. The augmented table has been shown in the truth table (10.3).

EN	B	A	Y_1	Y_{OR}
H	X	X	L	—
L	0	0	I_0	0
L	0	1	I_1	1
L	1	0	I_2	1
L	1	1	I_3	1

Truth table (10.3).

Thus, if $BA=00$, I_0 is selected at the output and if I_0 is connected to logic low, we will get low at the multiplexer output. Similarly, if we apply $BA=01$, I_1 is selected and if I_1 is connected to logic high, the

output will be at logic high, thus we can see that multiplexer behaves as OR gate. The Circuit diagram has been shown below: in figure (10.2).



(c)

OBJECTIVE: Construction of AND gate using Multiplexer and study its functional table.

THEORY: AND gate performs ANDing between two Boolean variables. These variables can be taken as A and B. This is also a basic logic gate and is functionally incomplete.

The truth table of AND gate and multiplexer has been augmented in truth table (10.4).

We can see that when $BA=00$, I_0 is selected and if we change the combination at the input for $BA=01$, $BA=10$, the output of logic gate is at logic zero, thus if we connect all the input i.e. I_0 ($BA=00$), I_1 ($BA=01$) and I_2 ($BA=10$) are connected to logic low and I_3 ($BA=11$) is connected to logic high, then the multiplexer output will behave as an AND gate as shown in the figure (10.3).

EN	B	A	Y_1	Y_{AND}
1	X	X	0	-
0	0	0	I_0	0
0	0	1	I_1	0
0	1	0	I_2	0
0	1	1	I_3	1

Truth table (10.4).

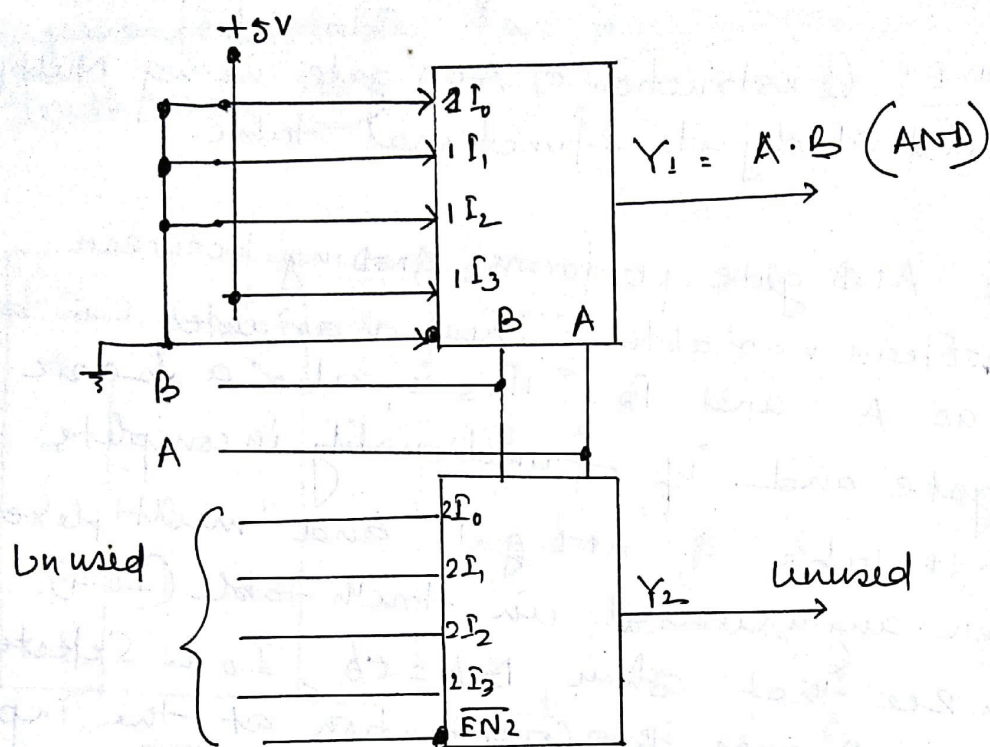


Figure (10.3) AND gate using multiplexer.

(c) OBJECTIVE: Construction of NOT gate using multiplexer

THEORY:

NOT gate is also a basic logic gate and it is having only one logic input. Since, we are using 4 to 1 multiplexer with two select lines and the NOT gate is having only one input, therefore we can use only one select line or short both the select lines to make it one.

If both the select lines are connected together only two combinations are possible i.e. $AB = 00$ or $AB = 11$ because if both the select lines

(7)

are Connected then, the Select lines becomes

$$A \cdot A = 00 \text{ and } A \cdot A = 11$$

when $AA = 00$, I_0 is Selected at the output

and when $AA = 11$, I_3 is Selected at the output of the multiplexer. I_2 & I_1 inputs are never Selected,

thus we can connect I_0 to Logic high and I_3 to Logic Low. Thus, when Select lines are

00 we get high at output and when it is

11 we get Low. Thus, it behaves as NOT gate.

The circuit of NOT gate using multiplexer as shown in figure (10.4)

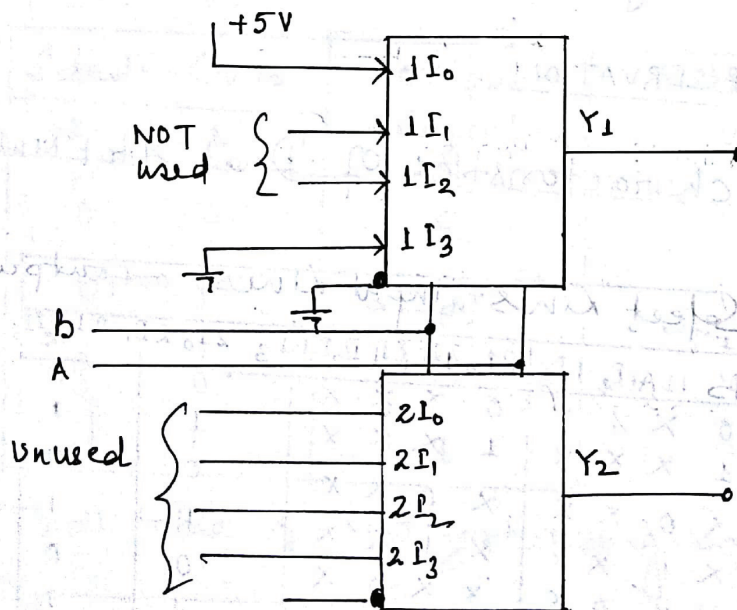


Figure (10.4), NOT gate using multiplexer.

APPARATUS REQUIRED:

- (i) Semiconductors integrated Circuits
 - (a) 74LS153 - Dual 4 to 1 multiplexer
 - (b) Light emitting diode (LED)
- (ii) Regulated 5 volts positive voltage power supply
- (iii) Bread board
- (iv) Jumpers or Connecting wires
- (v) Digital voltmeter
Range : 0-20volts dc
- (vi) Current limiting resistor 680Ω

EXPERIMENTAL OBSERVATION:

(a) Functional characteristics of Dual 4 to 1 Multiplexer

Enable		Select		Input 1				Input 2				output	
EN1	EN2	B	A	I ₀	I ₁	I ₂	I ₃	2I ₀	2I ₁	2I ₂	2I ₃	Y ₁	Y ₂
0	0	0	0	0	X	X	X	0	X	X	X	0	0
0	0	0	0	1	X	X	X	1	X	X	X	1	1
0	0	0	1	X	0	X	X	X	0	X	X	0	0
0	0	0	1	X	1	X	X	X	1	X	X	1	1
0	0	1	0	X	X	0	X	X	X	0	X	0	0
0	0	1	0	X	X	1	X	X	X	1	X	1	1
0	0	1	1	X	X	X	0	X	X	X	0	0	0
0	0	1	1	X	X	X	1	X	X	X	1	1	1

Note:

(b) Truth table of OR gate using multiplexer, dual 4 to 1 multiplexer: 74LS153

Select Lines		output	output in volts
B	A	Y_1	
0	0	0	
0	1	1	
1	0	1	
1	1	1	

(c) Truth table of AND gate using dual 4 to 1 multiplexer 74LS153

Select Lines		output	output in volts.
B	A	Logic (Y_1)	
0	0	0	
0	1	0	
1	0	0	
1	1	1	

(d) Truth table of NOT gate using dual 4 to 1 multiplexer

Select Line	Logical output (Y_1)	output in volts.
A		
0		
1		

DISCUSSION :

- (i) The Enable Lines of each multiplexer inside must be active ie must be Connected to Logic Low.
- (ii) The multiplexer before can it be used must have the Supply pins Connected to Regulated +5 volt dc power Supply.
- (iii) Wire used must have less thickness to avoid damage to the internal connectors of the bread board.
- (iv) The integrated circuits used must be properly checked for loose connection.
- (v) TTL IC used get affected by floating input, therefore none of the pin must be left open, if it requires low logic, then it must be grounded and if it require high logic, then it must be Connected to Supply (+Vcc).
- (vi) Light emitting diodes must be Connected via Current Limiting resistor.